State Machine Diagrams

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Working with State Diagrams

- Use Cases and Scenarios provides a way to describe system behaviour.
- ***Use Case*** – Typical interaction between a user an a computer system.
- ***Scenario*** – Instance of a Use Case
- ***Interaction Diagrams*** – Capture Scenarios.

  Shows object interactions arranged in time sequence.
- Some times it is necessary to look at the behaviour **inside an object.**

Working with State Diagrams

- As the system interacts with users and other systems,
  - The objects that make up the system go through necessary changes to accommodate the interactions.
- If you are going to model systems, you must have a mechanism to model change.
- One way to characterize change is to say that its objects change their state in response to events and to time.

Working with State Diagrams

Examples:

- When you throw a switch, a light changes its state from Off to On.
- After an appropriate amount of time, a washing machine changes its state from Washing to Rinsing.
- Hotel room changes its state to available, reserved and occupied.

- UML State diagram captures these kinds of changes.

Working with State Diagrams

- UML *State Transition Diagrams* shows:
  - Life history showing the different states of a given object.
  - The events or messages that cause a transition from one state to another.
  - The actions that results from a state change.
- *State Diagrams* are created only for classes with <u>significant</u> dynamic behaviour.

  eg. ***Hotel Room*** in a Hotel Reservation System

Modeling Dynamic Behaviour

- Interaction diagrams can be studied to determine the dynamic objects.
  - Objects receiving and sending many messages.
- If you have an attribute called *status*.
  - This can be a good indicator of various states.

States

Condition or Situation during the life of an object

- eg. HotelRoom object can be in one of the following states.
  - Occupied, Available, Reserved
- eg. Course object (in a course registration system) can be in one of the following states.
  - Initialization,Open, Close,Cancel

UML Notation for a State

State Transitions

- A State Transition represents a change from an originating state to a successor state.
- An action can accompany a state transition.
- A State Transition is represented by an arrow that points from the originating state to the successor state.

UML Notation for State Transition

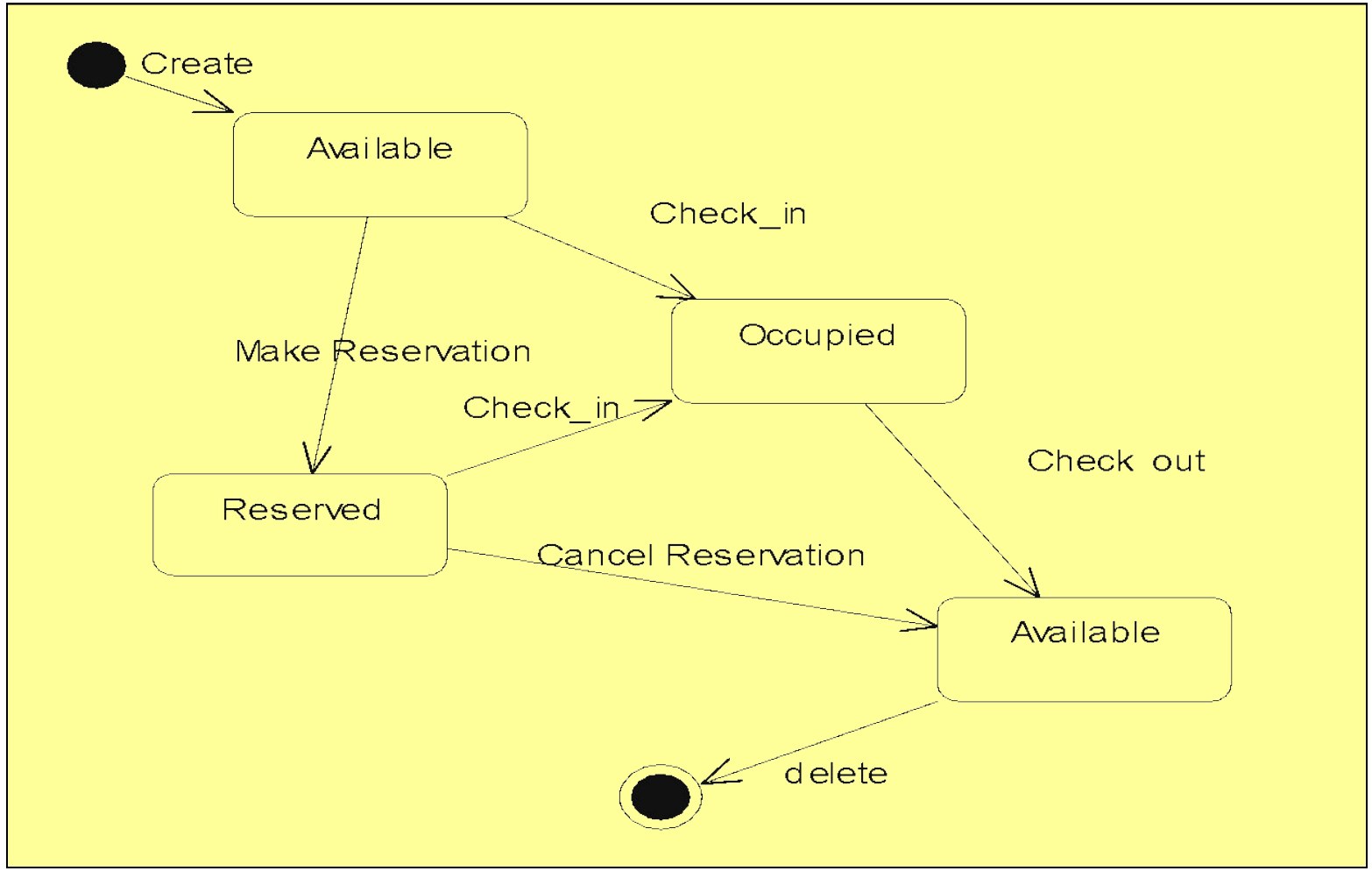
Special States

- There are two special states that are added to the state transition diagram.
- **Start** state – Each diagram must have one and only one start state.
- **Stop** state – An object can have multiple stop states.

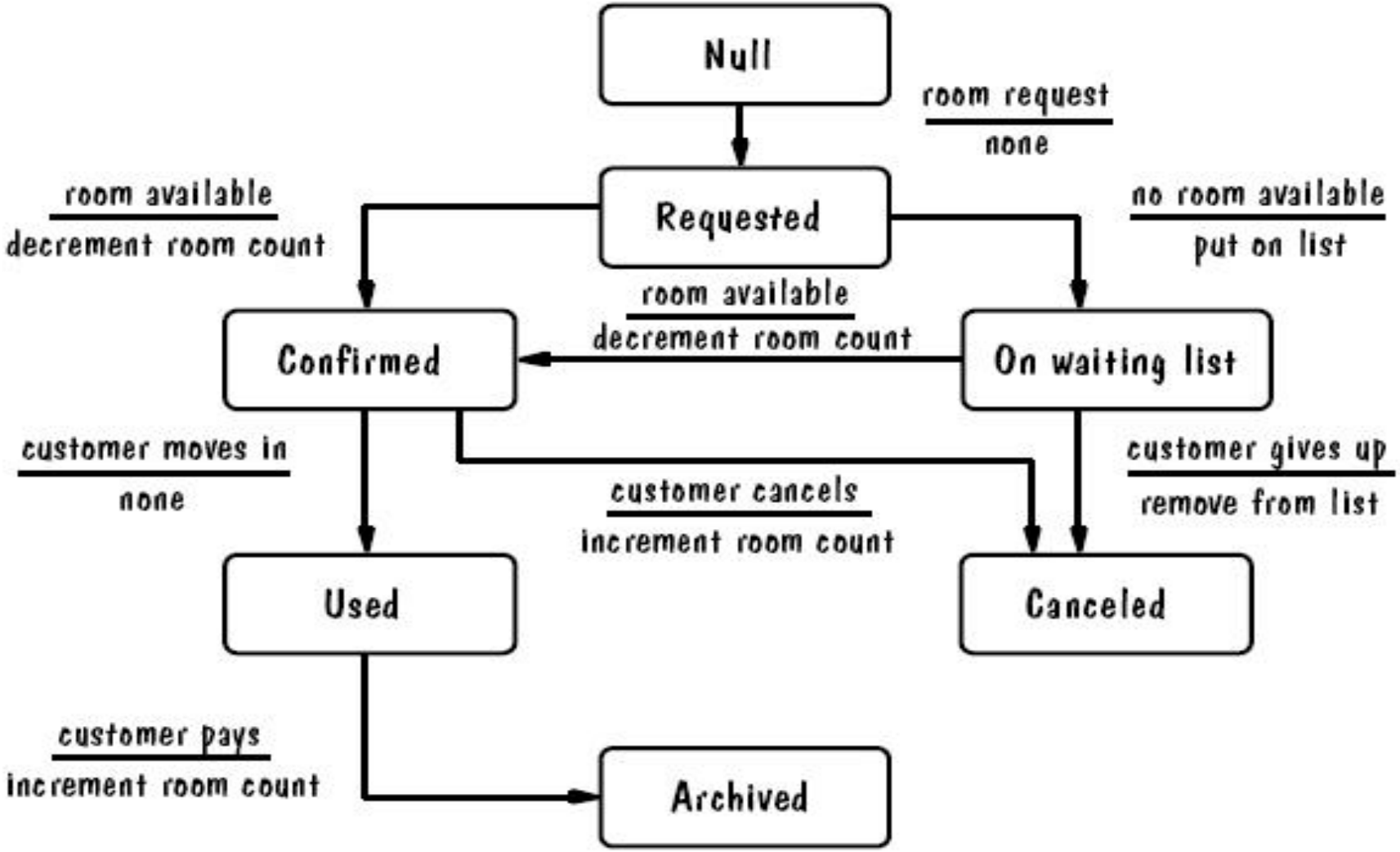
Start State

Stop State

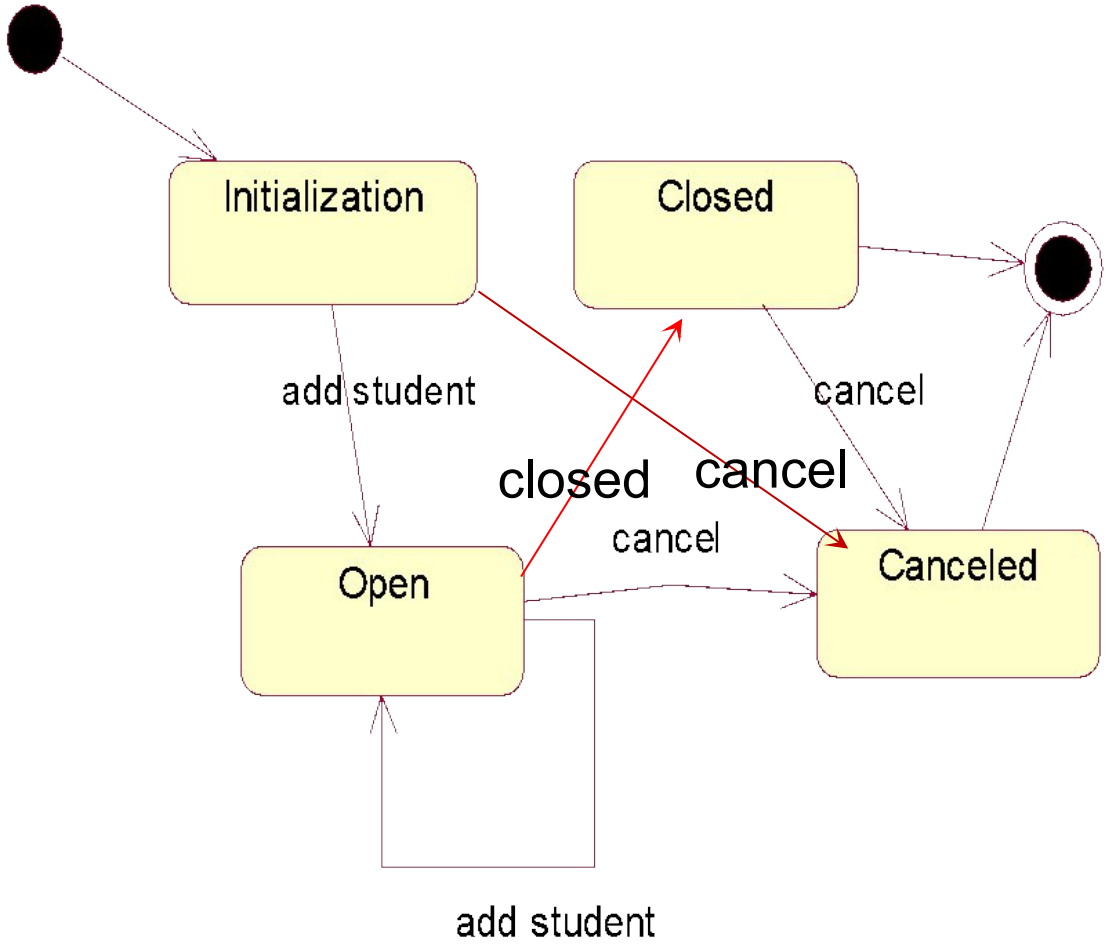
State Transition Diagram –Hotel Room Class



Hotel Room – Another example



State Transition Diagram– Course Class



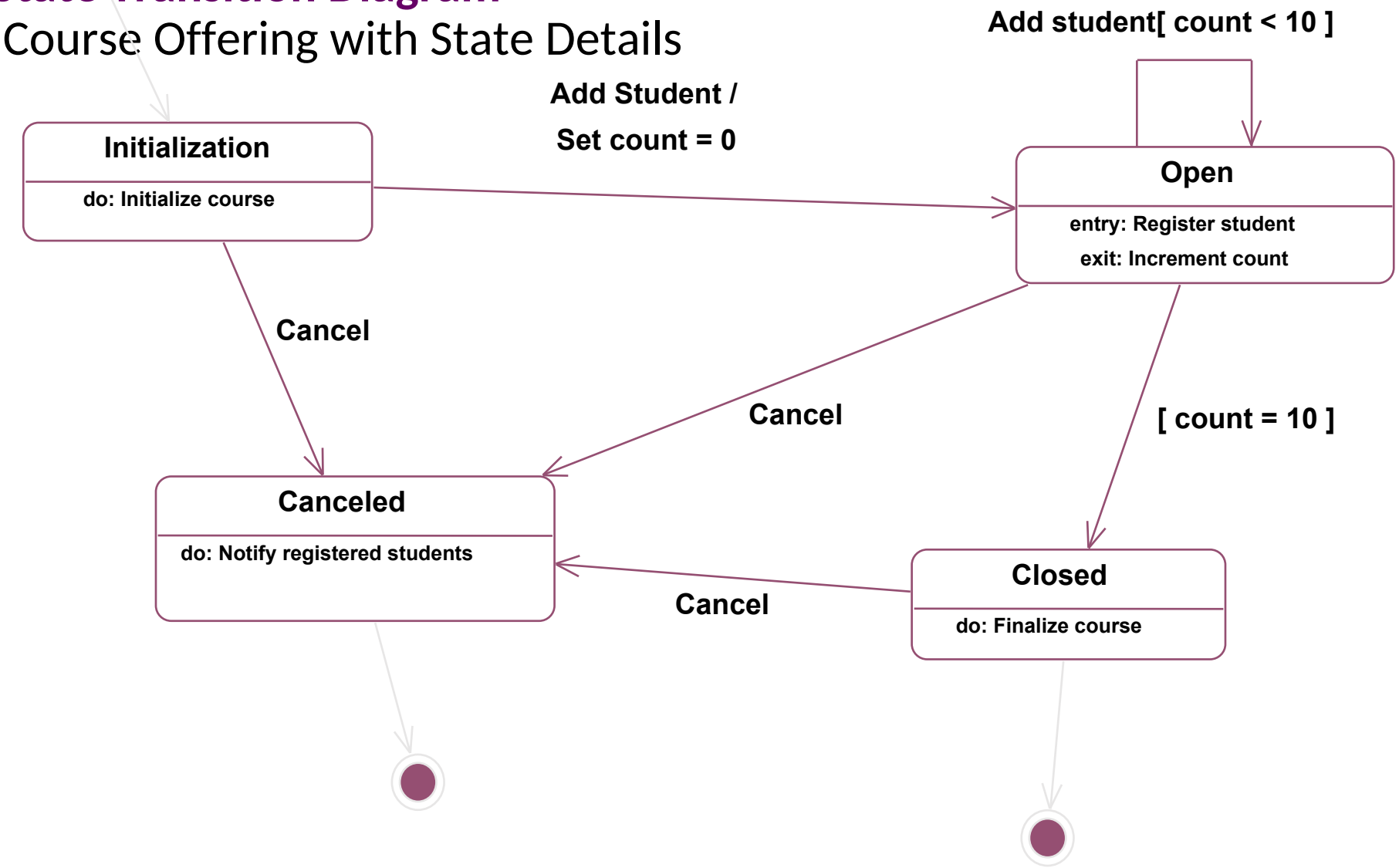
State Transition Details

- A State Transition may have the following associated with:
  - an action and/or
    (behaviour that occurs when the state transition occurs.)
  - a guard condition
    (allows state transition only if it is true.)
- A State Transition may also trigger an event

A message that is sent to another object in the system.

State Transition Diagram

Course Offering with State Details



State Details

- ***Activity*** : behaviour that an object carries out while it is in a particular state.
  - An activity is shown inside the state itself, preceded by the word *do* and a colon.
- **Entry Action :**
  - Behaviour that occurs while the object is transitioning into the state.
  - Shown inside the state, preceded by the word *entry* and colon.

State Details cont…

- **Exit Action :** occurs as part of the transition out of a state.
  - Shown inside the state, preceded by the word *exit* and colon.
- The behaviour in an activity, entry action, or exit action can include sending an event to some other object.

State Details con…

- In this case, the activity, entry action, or exit action is preceded by a **^**

  Do:^Target.Event(Arguments)

  Target - object receiving the event

  Event - message being sent

  Arguments – parameters of the message being sent

  Eg.

  Do:^CourseRoster.Create

UML 2.0 State Diagrams

- UML 2.0 has added some new state relevant symbols called **connection points.**
- They represent points of entry into a state or exists out of a state.
- Lets look at the different state of a book in a library.

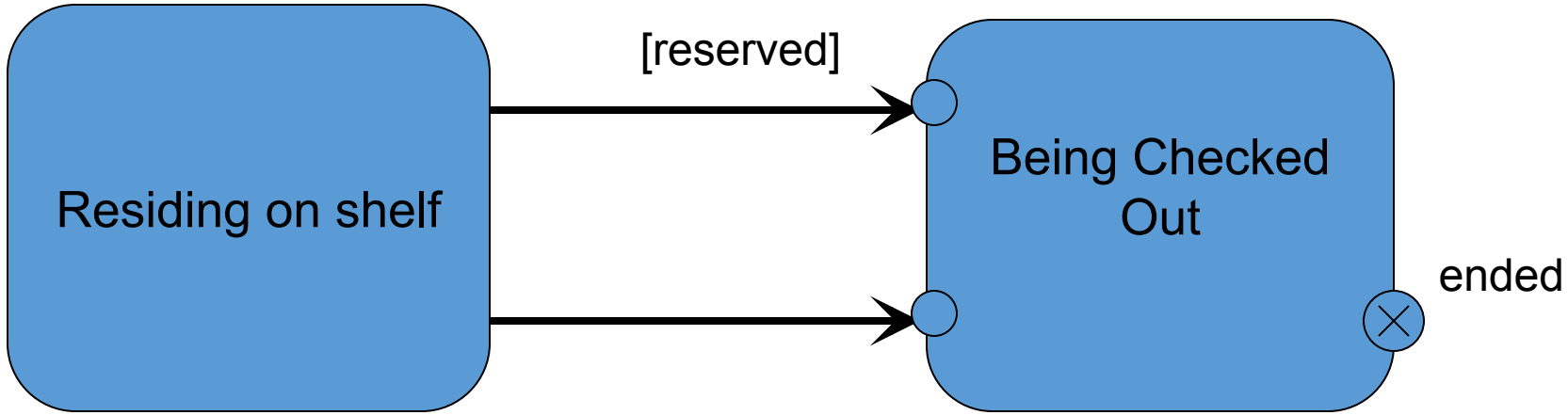
UML 2.0 State Diagrams

- At first the book is residing on a shelf.
- If a borrower has called in to reserve the book, a librarian retrieves the book and brings it into the state of *“Being checked out”*.
- If a borrower comes to the library, browses through the shelves, selects the book, and decides to borrow it.
- Again it enters into the state of *“Being checked out”*, but in a different way.

UML 2.0 State Diagrams

- You can think of each way of getting to the Being-checked-out state as going through a separate **entry point**
- Suppose the borrower is trying to borrow more than the allotted limit or has number of unpaid fines.
- If that is the case the book abruptly exits via an **exit point,** from "*Being-checked-out*" *state*

Entry points and exit point in a UML state diagram



Why are State diagrams important?

- They model the changes that just one object goes through.
- They help analysts, designers, and developers understand the behavior of the objects in a system.
- A Class diagram and an object diagram show only static aspects of a system. They do not show the dynamic details of the behaviors.

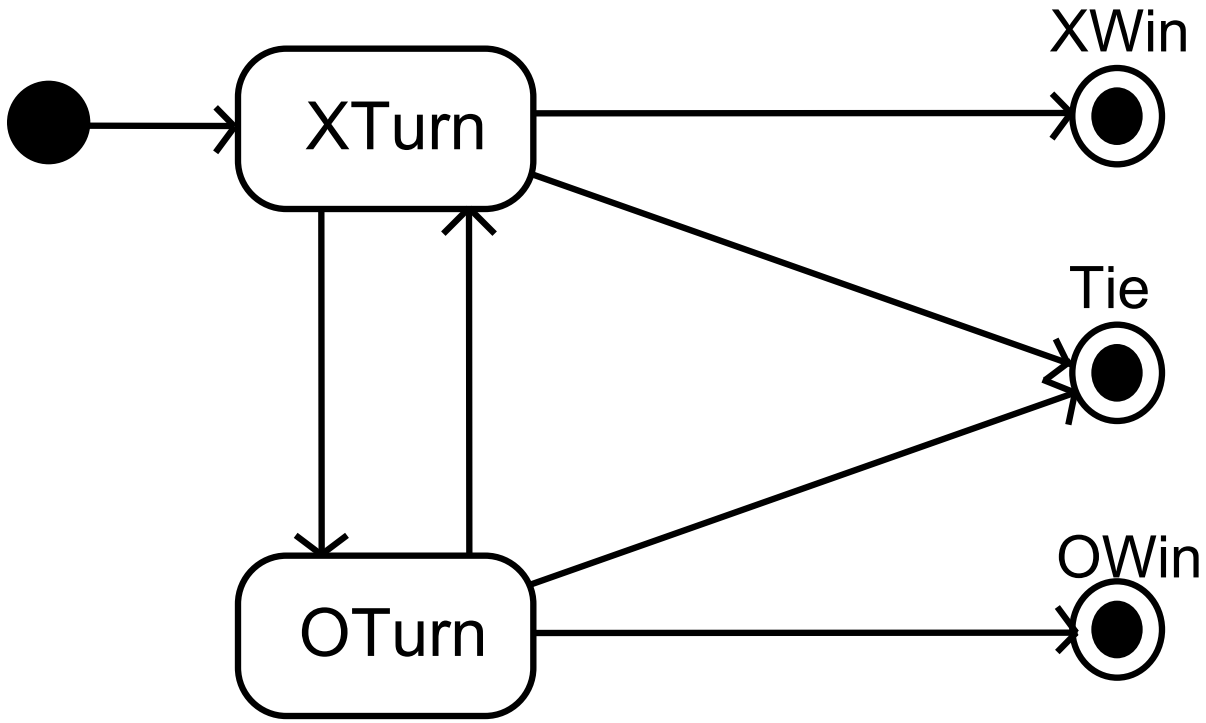
Why are State diagrams important?

- Developers, in particular, have to know
  - how objects are supposed to behave

    because they have to implement these behaviors in software.
  - It is not enough to implement only objects.
  - Developers have to make that object do something.
- State diagrams ensure that they won't have to guess about what the object is supposed to do.

State diagrams – an example

- tic-tac-toe game



States

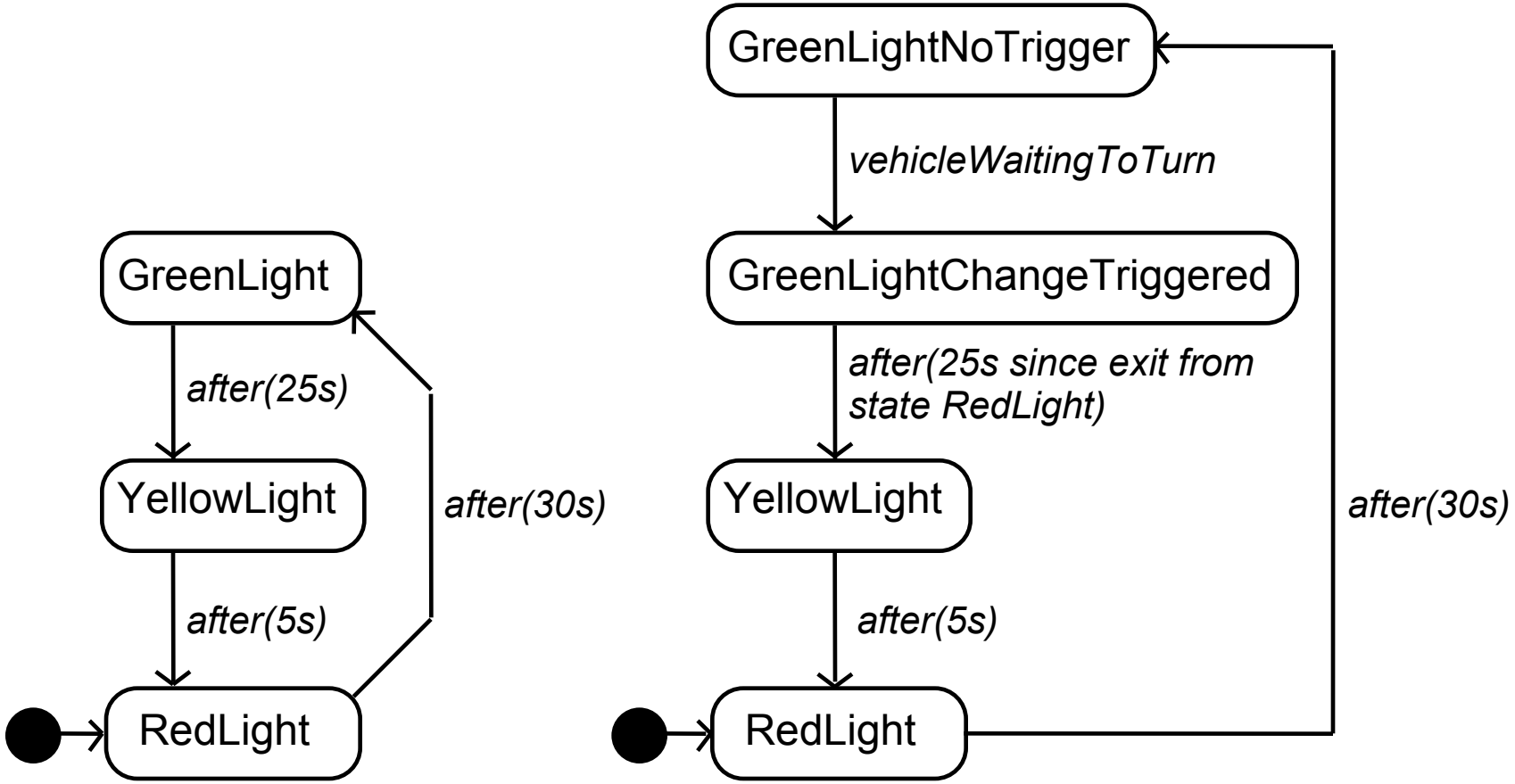
- At any given point in time, the system is in one state.
- It will remain in this state until an event occurs that causes it to change state.
- A state is represented by a rounded rectangle containing the name of the state.
- Special states:
  - A black circle represents the *start state*
  - A circle with a ring around it represents an *end state*

Transitions

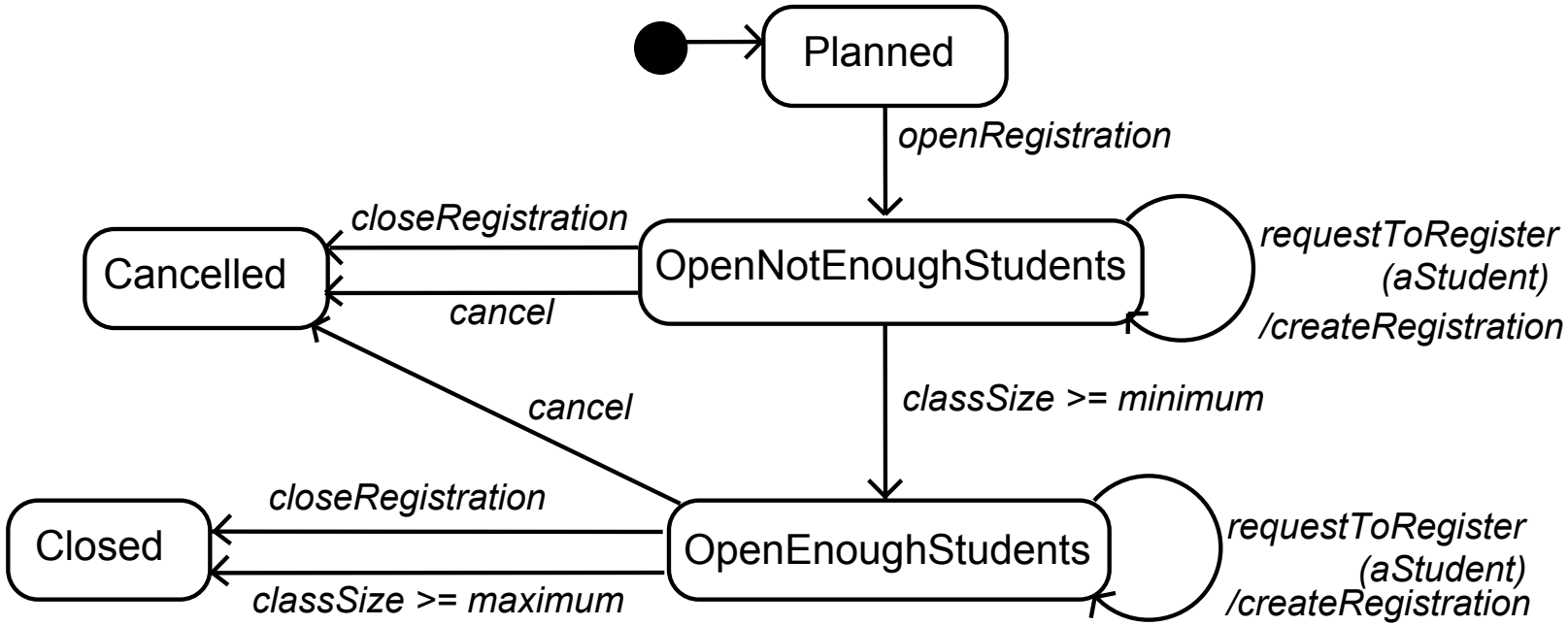
- A transition represents a change of state in response to an event.
  - It is considered to occur instantaneously.

- The label on each transition is the event that causes the change of state.

State diagrams – an example of transitions with time-outs and conditions



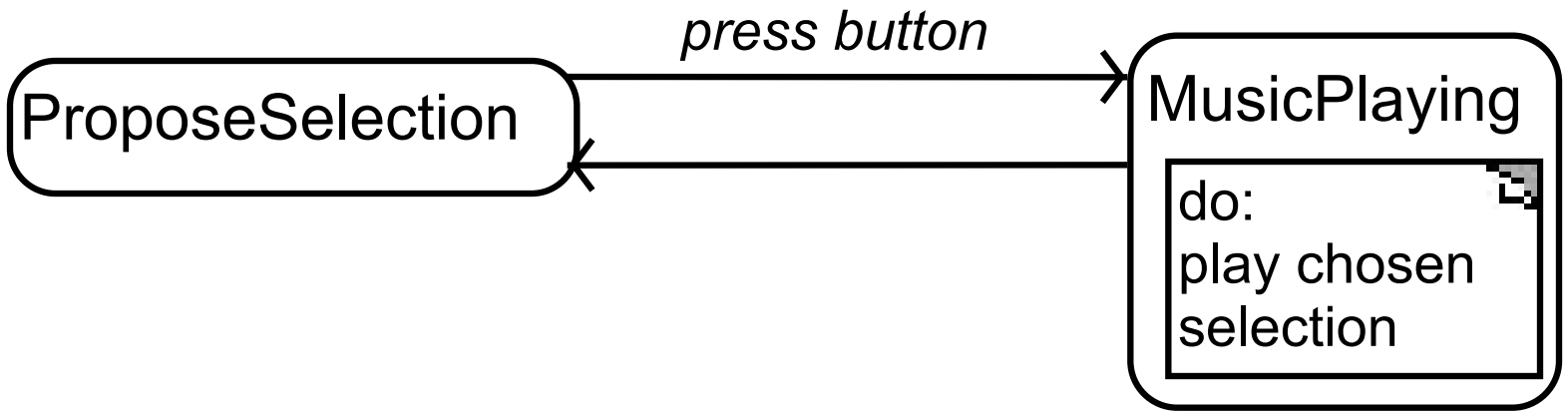
State diagrams – an example with conditional transitions



Activities in state diagrams

- An *activity* is something that takes place while the system is *in* a state.
  - It takes a period of time.
  - The system may take a transition out of the state in response to completion of the activity,
  - Some other outgoing transition may result in:
    - The interruption of the activity, and
    - An early exit from the state.

State diagram – an example with activity

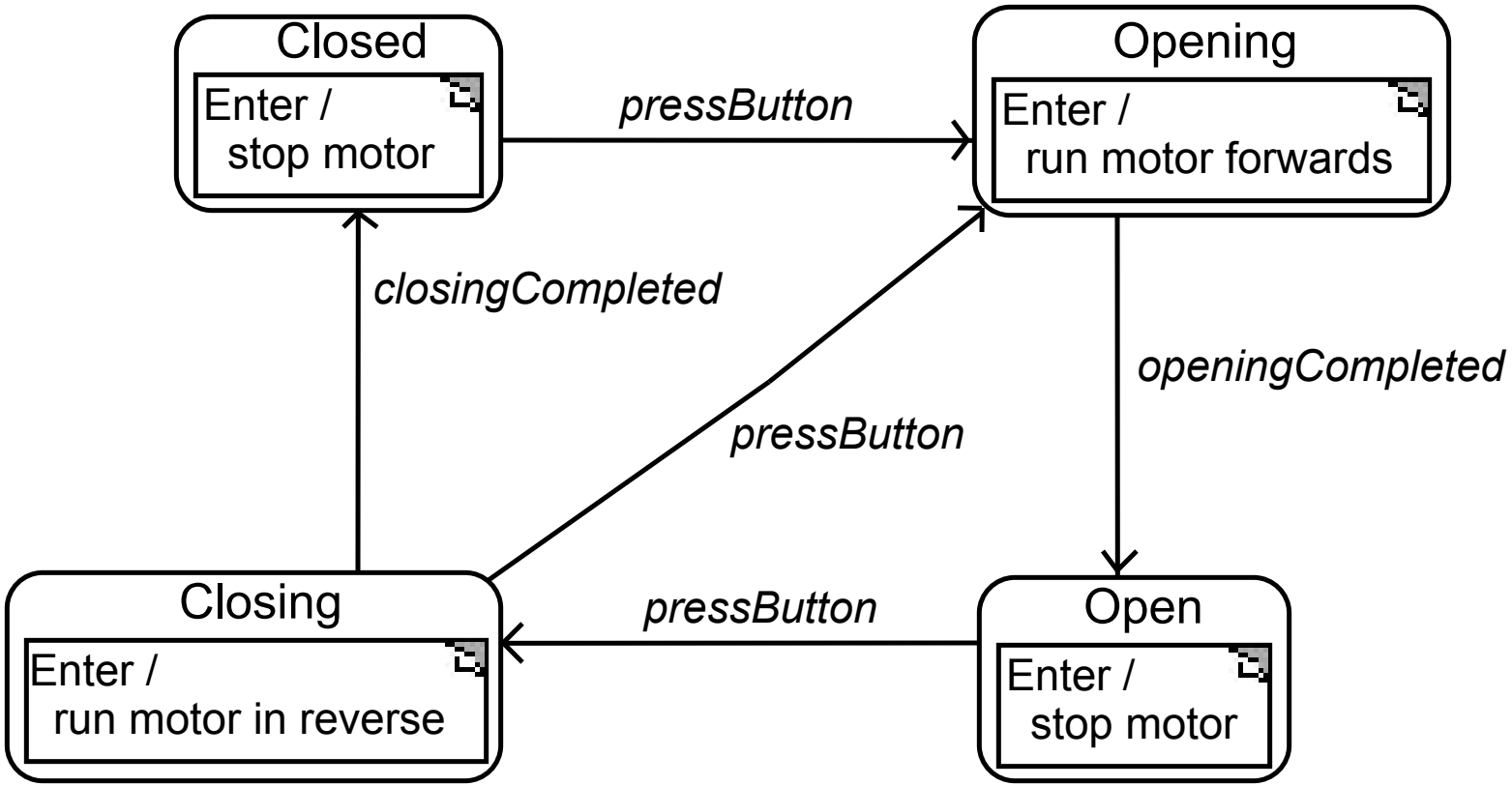


Actions in state diagrams

- An *action* is something that takes place effectively *instantaneously*
  - When a particular transition is taken,
  - Upon entry into a particular state, or
  - Upon exit from a particular state

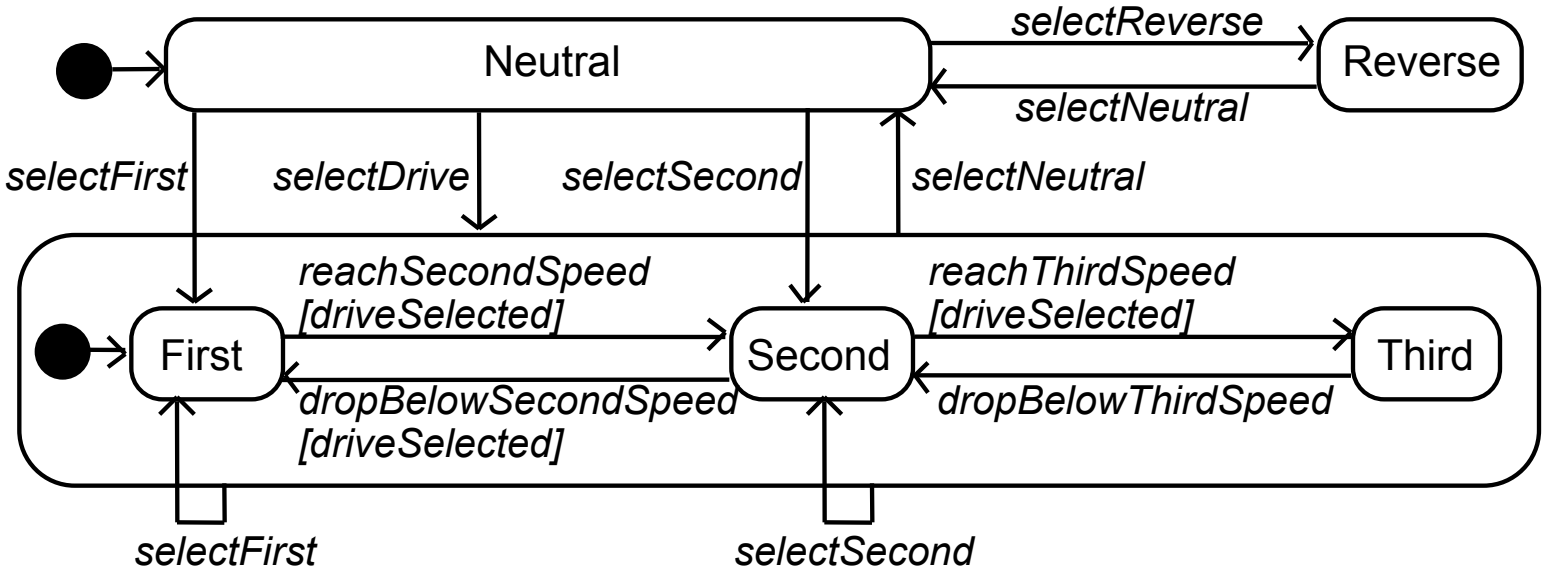
- An action should consume no noticeable amount of time

State diagram – an example with actions



Nested substates and guard conditions

- A state diagram can be nested inside a state.
  - The states of the inner diagram are called *substates*.



Activity Diagrams

- An *activity diagram* is like a state diagram.
  - Except most transitions are caused by *internal* events, such as the completion of a computation.

- An activity diagram
  - Can be used to understand the flow of work that an object or component performs.
  - Can also be used to visualize the interrelation and interaction between different use cases.
  - Is most often associated with several classes.

- One of the strengths of activity diagrams is the representation of *concurrent* activities.

Q & A